Aiden Cox

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Education

Wright State University **August 2023 – May 2026**
Bachelor of Science in Computer Science | GPA: 3.9 / 4.0 *Fairborn, Ohio*

Experience

Booz Allen Hamilton **May 2026 – Present**
Software Engineer | Active Secret Clearance (TS/SCI In Progress) *Beavercreek, Ohio — Wright Patterson Air Force Base*

- Develop and maintain software for defense simulation and modeling systems in a classified environment supporting U.S. Air Force mission requirements.
- Contribute to simulation tooling using C++ within AFSIM (Air Force Simulation) and supporting frameworks including Wizard and Mystic.
- Work within a secure, access-controlled development environment adhering to DoD software and security standards.

Teaching Assistant **August 2024 – May 2026**
Operating Systems, Concepts, and Usage - Lab Leader *Fairborn, Ohio*

- Lead a Lab of 40-50 students teaching core concepts of Computer Science. Duties included assisting students during the lab session by giving students one-on-one attention. Help grade and provide feedback on lab assignments. Holding two in-person office hours a week, and aided in monitoring questions and conversations in the Discord channel.
- Students used Visual Studio Code, Terminal, WSL2 + Ubuntu, Bash, and Git.

Wright-Patt Credit Union **May 2025 – December 2025**
Data Analytics / Automation Intern *Fairborn, Ohio*

- Supported business operations by streamlining data workflows and automating repetitive tasks in Excel using VBA, resulting in increased efficiency and reduced errors. Developed and maintained interactive dashboards in PowerBI to visualize key performance indicators, trends, and actionable insights. Collaborated with team members to gather requirements, clean datasets, and deliver analytics solutions that informed strategic decisions.
- Using Excel, Visual Studio Code, Python, Jupyter Notebook, and PowerBI to complete tasks.

Projects

JPL Construction App - BufferFlushed Labs LLC | *Expo, Python, FireStore, TypeScript* **May 2025 - Current**

- Collaborated cross-functionally to develop a scalable, mobile-responsive Job Finance App aimed at streamlining business operations for invoicing, client tracking, and financial reporting.
- Engineered dynamic data models for job, client, and invoice management using real-time cloud storage and integrated automated PDF generation and email dispatch functionality.
- Designed and implemented a secure customer portal to manage payments and client interactions, improving user experience and reducing administrative workload.
- Integrated planned enhancements including data-driven financial dashboards, automated recurring billing, and animated UI transitions to boost engagement and usability.
- Utilized Agile development practices, version control, and team-based planning tools to ensure efficient project delivery and high-quality codebase maintenance.

Senior Design Project Lead | *Python, OpenCV, Bash, RFM69, HTML/CSS/JS* **August 2025 - May 2026**

- Led a 5-person multidisciplinary team through all phases of senior design project, coordinating computer science and electrical engineering efforts to deliver an autonomous secure data transmission system under $300 budget constraint.
- Architected and implemented end-to-end pipeline automation using Bash orchestration, integrating computer vision (OpenCV), AES-256 encryption, RF wireless transmission (RFM69 433MHz), and web-based visualization with two-factor authentication.
- Designed covert RF communication protocol capable of transmitting encrypted payloads over 5+ meters within 600-second window, implementing custom chunking and checksum verification for data integrity.
- Developed locally-hosted web interface with multi-factor authentication (password + personal access token) for tiered access control, featuring dynamic image loading and embedded video content.
- Established Git-based version control workflow, conducted code reviews, and maintained comprehensive documentation across multiple repository modules, ensuring seamless team collaboration.

Leadership / Extracurricular

* Tau Beta Pi Engineering Honor Society Treasurer
* ACM Programming 2024 and 2025 Judge
* Computer Engineering and Computer Science Alumni Society Board Member